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Module 2: Introduction

Lesson 1: Estimating Probabilities and Posterior Distributions

Reading: Estimating Probabilities and Posterior Distributions Reading 10 min

Video: Binomial and Posterior Distributions Pt. 1 5 min

Video: Binomial and Posterior Distributions Pt. 2 5 min

Video: Binomial and Posterior Distributions Pt. 3 5 min

Video: Binomial and Posterior Distributions Pt. 4 5 min

Video: Binomial and Posterior Distributions Pt. 5 5 min

Video: Binomial and Posterior Distributions Pt. 6 5 min

Practice Assignment: Estimating Probabilities and Posterior Quiz 13 min

Lesson 2: Summarizing Posterior Inference and Prior Distributions

Lesson 3: Normal Distribution and Other Single-Parameter Models

Module 2 Summative Assessment

Module 2 Summary

Your grade: 100%

Your best: 100% • Your highest: 100%

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Estimating Probabilities and Posterior Quiz

Instructions

Review Learning Objectives

1. What happens to the posterior distribution as the sample size increases? 1 / 1 point

☒ It becomes more influenced by the data.

☐ It becomes less variable as the sample size increases.

☐ It remains equally influenced by the prior distribution.

☐ It becomes more influenced by the prior distribution.

Correct

Assignment details

Correct! As the sample size increases, the posterior distribution is more influenced by the data than the prior distribution. Submitted Nov 18, 9:20 PM CST Attempts 0/100000

Try again

2. What is the prior distribution assumed in the example of estimating the probability of female births? 1 / 1 point

☒ Uniform distribution

☐ Normal distribution

☐ Gamma distribution

☐ Exponential distribution

Correct

Correct! A uniform prior distribution is assumed in the example, which results in the posterior distribution being a Beta distribution.

3. Suppose you are tracking births in an isolated community and out of the 100 births you observe, 49 are female. What is the expected value of θ ? 1 / 1 point

☐ 48

☐ Unable to answer the question

☒ 49

☐ 59

Correct

Assuming a uniform distribution for θ , we know the posterior distribution is the Beta distribution based on the supplied data. What is this distribution's expected value?